



\$ whoami

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EXPERIENCE

NVIDIA Corporation – *developer.nvidia.com/cuda-downloads*

Jan 2017 – present

Senior Systems Software Engineer V

2024 – present

- Acting manager for a medium-sized team of engineers – oversaw sprint schedule, bug scrubbing, prioritization calls, status roll-up, escalation and execution tasks. Conducted emergency meetings to mitigate day-by-day slip and future corrective action.
- Team lead and 1-on-1 mentoring for new hires getting them up to speed on design, build, and sustain of packaging workflows.
- User Experience group leader improving internal user story workflows, architectural review, and reduction of technical debt.
- PIC (Pilot in Charge) for initiative NVIDIA Open GPU Kernel Modules by default, new nvidia-open meta-package upgrade path, ABI behavior changes to support business logic within distro-specific constraints, and a major documentation overhaul.
- Installer advocate, putting in motion initiatives including revamped documentation with separate guides for CUDA and Driver installation, improved discover-ability for Datacenter Drivers with download page, lightning talks & AMAs (Ask Me Anything).

Senior Systems Software Engineer IV

2021 – 2024

- Embarked on CUDA-X installer refresh of cuDNN 9 to improve user experience moving CUDA variant from version to name.
- Lead the charge for migrating from a monolithic Perforce codebase to componentized git repos improving branching and OSS.
- Designed distro-specific packaging of NVIDIA Open GPU Kernel Modules for RHEL (kmod-nvidia-open-dkms), Ubuntu (nvidia-kernel-open), Fedora (nvidia-driver:515-open), SUSE (nvidia-open-gfxG05), and Debian (nvidia-kernel-open-dkms).
- Implemented a toggle in the ncurses CUDA runfile using C++ `#include` guards to opt into new `-m=kernel-open` module flavor.
- Shepherded GPG key rotation, signing 1TB of shipped artifacts in 25 repos, enabling HSM pipelines, cuda-keyring package, ephemeral keys for reduced surface area with local repo installers, and customer interaction for migration to the new pubkey.
- Principal engineer for redistributable “binary archives” initiative, delivering tarball and zip archives accompanied with JSON manifests per release indicating LICENSE, checksums, etc. for use with 3rd party CI/CD build systems and packaging.

Senior DevOps Engineer III

2020 – 2021

- Spearheaded GitHub open source package template initiative, rebuilding over a dozen internal repositories with interleaved git history, using mailbox patches to redact sensitive commit messages and correspondence with community contributors.
- Coordinated with stakeholders across timezones to ship several new products on tight deadlines, developing custom installer solutions, and ensuring best practices followed using semantic versioning and identifying package dependencies.
- Garnered a reputation for distilling complex software ecosystems into easy to understand presentations and visual analysis.
- PIC for precompiled driver kernel module packaging solution, co-developed with Red Hat engineers to improve GPU cloud server instances, providing hardening and UEFI secure boot support. Modular streams to support user choice to lock updates to a specific branch, stay up-to-date with branch and kernel, or fallback to DKMS. End-to-end automated CI/CD with 24h SLA.
- Calendar-based, ShipIt Scheduler, provides program management a centralized dashboard to plan automated deployments with product specific approval and Slack progress notifications. A train model prevents metadata collisions and CDN overhead.

Systems Software Engineer II

2018 – 2020

- Built ShipIt, an in-house SaaS (Software as a Service) full-stack webUI for managing product releases using a combination NodeJS and Perl backend. Turn-key ease of use to provide management, a reliable data-driven interface populated by JSON.
- Shaped Kitmaker, a containerized approach to CI/CD, supporting many products and architectures, by converting monolithic builds into componentized fragments using Artifactory, Vault, and Jenkins to produce customizable and reproducible pipelines.
- Envisioned Project Triforce, an OverlayFS based state machine for managing repository metadata as atomic transactions. Significant quantitative benefits for CPU, disk, and network resources using a three-pronged approach & incremental snapshots.
- Coordinated closely with QA engineers to provide heads-up on upcoming changes, technical consultations and tooling.
- Documented detailed end-to-end pipeline process from source code check-in to release candidate using visuals and diagrams.
- Cross-team maintainer for the NVIDIA Machine Learning repository, handling the logistics of interlaced product releases.

Systems Software Engineer I

2017 – 2018

- Designed AUTOPKG, a fully automated continuous deployment pipeline for packaging of the CUDA SDK for the plethora of supported distributions, platforms and installation formats. Layered architecture design contributed increased release cadence.
- Deployed a transparent squid proxy to rewrite public URLs for seamless cross-platform internal testing with one-line. Mechanism has saved time and provided assurance that network installers are bit-identical during testing and for public release.
- Maintainer for CUDA and driver installation packages using Perforce and git for Fedora, RHEL/CentOS, SUSE, Ubuntu, L4T (Linux4Tegra), Windows and MacOS. Strong expertise with rpm build and .SPEC files, deb uild and control/rules files.

OPEN SOURCE

apacman – Lead Maintainer of Community Project

github.com/oshazard/apacman 2013 – 2016

- Took over reins of popular BASH-based, package manager for ArchLinux User Repository (AUR) PKGBUILD templates
- Created a configuration file, many new features, command-line (CLI) flags, added Bats unit tests and wrote manpage docs

JustBrowsing Linux – Founder and Project Manager

justbrowsinglinux.com 2013 – 2016

- Developed ChromeOS-like immutable kiosk Linux distro for web browsing with lock screen webcam intrusion detection.
- All-in-one settings control panel with persistent config file; wrote calculator, timers webapps and pyGTK keyboard applet.

TECHNICAL SKILLS

20+ years of GNU/Linux experience

- Source packaging, Makefiles, porting and maintenance

Command-line and scripting proficiency

- awk/grep/sed, busybox, chroot, ssh, apt-get, dnf, zypper, vault

Writing Documentation

- manpages, draw.io, dot, UML, LaTeX, markdown, rST/Sphinx

Development Tools

- Android Studio, Bats, Eclipse, git, gcc, p4, meld, Scratchbox

Programming Languages

- Bash, Perl, JavaScript, Go, Python, Groovy, Java, R, C++

Full-stack Web: Back-end and Front-end

- NodeJS, Socket.io, CSS/HTML, Bootstrap, jQuery, XML

Database and Graphics

- MySQL, SQLite, JSON, YAML, Artifactory, ImageMagick

Containers, Virtualization, and Deployment

- docker/podman, nspawn, QEMU, VirtualBox, virsh, Jenkins

PROJECTS

CUDA repo management – Lead Developer 2018 – present

- Managing Debian and RPM repositories with many packages.
- Scripts append-only, OverlayFS, mirroring, metadata validation

RHEL precompiled – Co-developer 2018 – 2020

- Lead NVIDIA engineer in collaboration with Red Hat engineer
- Precompiled kernel module (kmod) packages: driver @ kernel
- Plugins for dnf and yum package managers for .ko files

GuessWhat – Core Developer Spring 2016

- Online Pictionary game with real-time multiplayer and chat
- Node.js and JSON API to fetch words, stored in Redis DB

TimeMe – Lead Developer Spring 2012

- CSV-based task tracking application written in Java SWT
- Developed with Eclipse IDE, git, SRS and UML diagrams

webGPS – Developer Mid 2011

- Parse GPS coordinates from Palm webOS phone via USB
- PHP shim client-server model mapped APs with gpsd + kismet

NVIDIA driver templates – Maintainer 2017 – 2024

- Maintainer for six sets of distro-specific driver templates.
- Collaboration with open-source packaging community

NVIDIA ShipIt – Lead Developer 2017 – present

- Turn-key full-stack webUI for managing product releases
- JSON data-driven interface populated via Artifactory AQL
- Global terminal using socket.io, Node.js, and Perl back-end

pyCheckmate – Developer Fall 2015

- Python chess end-game AI, 5-ply minimax α - β pruning
- Wrote user interaction and QA test prep for tournament

PrinterPi – Project Manager Q3 2012

- iPhone printing bridge Raspberry Pi CUPS + hostapd WiFi
- QEMU user mode to run x86 binary drivers on ARM SoC

Maemo – Community Member Mid 2009

- Developed shell scripts for Debian-based Nokia N810 tablet
- Ported cross-compile Maemo Extras packages in Scratchbox

EDUCATION

Advanced Software Systems Graduate Program – Stanford University

2021 – TBD

- Advanced Topics in Operating Systems
- Mining Massive Data Sets

Bachelor of Science in Compute Science – California State University, Fullerton

Graduated 2016

- Data Structures
- Software Engineering
- Open Source Software
- Algorithms
- Network Security
- Web App Design

LEADERSHIP & PRESENTATIONS

99 Package Problems @ SCaLE 22x 2025

- Presented common RPM & Debian package manager errors

CUDA Packaging BoF (Birds of a Feather) 2023

- Discussion with Nixpkgs, Spack, Pixi at PackagingCon Berlin

Precompiled Kernel Modules & Modularity 2020

- Presented at GPU Technology Conference (GTC) in the Fall

Kitmaker Connect With Experts 2025

- AMA panel for CUDA Installer team

Triforce Repository Management 2021

- Presented at the first annual PackagingCon

Simplifying NVIDIA Driver Deployments 2020

- Presented at Red Hat Summit conference

CERTIFICATIONS & REPUTATION

NVIDIA Open GPU kernel modules by default 2024

- Blog: nvidia-open meta-packages

Streamlining NVIDIA Driver Deployment 2020

- Blog: RHEL modular streams for precompiled kmod packages

Accelerated Computing with CUDA C/C++ 2017

- Completed fundamentals course covering DIGITS

Applied CS with Android by Google 2016

- StackOverflow top 6% contributor in SuperUser

Updating the CUDA Linux GPG Repository Key 2022

- Blog: NVIDIA repo signing key migration & mitigation

Deep Learning Jetson Robotics Lab @ GTC 2018

- Robotics lab assistant image recognition, machine learning

Linux User Group @ CSUF 2014 – 2016

- Organized events and workshops for students and faculty

JustBrowsing in LinuxUser (DE) magazine 05.2014

Specialization in C++ / Achievement in CIS 2013